

COSC 594: Interplay Between RL and GenAI

Homework 1

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This homework is due on **9/17, 11:59pm**. Write your solution in Latex and submit it on Canvas.

1. (Policy gradient)

- (a) i. (1pt) Recall the discounted value function and the state visitation distribution,

$$V^\pi(\rho) := \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid s_0 \sim \rho \right]$$

$$d_\rho^\pi(s) = (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t \Pr(s_t = s \mid \pi).$$

Show the ‘linear’ form of the value function $V^\pi(\rho)$,

$$V^\pi(\rho) = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_\rho^\pi(\cdot), a \sim \pi(\cdot \mid s)} [r(s, a)]$$

Solution:

- ii. (1pt) Assume that $\sum_a \pi_\theta(a \mid s) = 1$. Derive the advantage version of policy gradient,

$$\nabla_\theta V^{\pi_\theta}(\rho) = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_\rho^{\pi_\theta}(\cdot), a \sim \pi_\theta(\cdot \mid s)} [A^{\pi_\theta}(s, a) \nabla_\theta \log \pi_\theta(a \mid s)].$$

Solution:

- (b) i. (1pt) Consider a general policy $\pi(\cdot | s): S \rightarrow \Delta_A$, and the mirror ascent policy update,

$$\pi^{(t+1)}(\cdot | s) = \operatorname{argmax}_{\pi(\cdot | s)} \langle Q^{\pi^{(t)}}(s, \cdot), \pi(\cdot | s) - \pi^{(t)}(\cdot | s) \rangle - \frac{1}{\eta} D_{\text{KL}}(\pi(\cdot | s) \| \pi^{(t)}(\cdot | s))$$

for all $s \in S$. Show that it is equivalent to the multiplicative weights update,

$$\pi^{(t+1)}(a | s) = \pi^{(t)}(a | s) \frac{\exp\left(\frac{\eta}{1-\gamma} Q^{\pi^{(t)}}(s, a)\right)}{Z^{(t)}(s)}$$

for all (s, a) , where $Z^{(t)}(s) := \sum_a \pi^{(t)}(a | s) \exp\left(\frac{\eta}{1-\gamma} Q^{\pi^{(t)}}(s, a)\right)$.

Solution:

- ii. (2pts) Prove the optimality gap of the T th policy iterate in running mirror ascent policy update,

$$V^*(\rho) - V^{(T)}(\rho) \leq \frac{1}{\eta T} \log |A| + \frac{1}{(1-\gamma)^2 T}.$$

Solution:

2. (Variational diffusion model)

- (a) A diffusion model consists of two key processes: a forward process and a backward process. The forward process begins with a data point x_0 , and generates latent variables $\{x_t\}_{t=1}^T$ from $t = 1$ to T . The joint distribution of latent variables results from conditional probabilities $q(x_{1:T} | x_0) = \prod_{t=1}^T q(x_t | x_{t-1})$, where $q(x_t | x_{t-1})$ is the distribution of latent variable x_t conditioned on the previous latent x_{t-1} , which is typically given by a Gaussian distribution. The backward process begins with the latent x_T sampled from the standard Gaussian $p(x_T) = \mathcal{N}(x_T; 0, I)$, and decodes latent variables from $t = T$ to $t = 0$ with a joint distribution,

$$p(x_{0:T}) = p(x_T) \prod_{t=1}^T p(x_{t-1} | x_t).$$

Recall the KL divergence of distribution q from distribution p by $D_{\text{KL}}(q \| p) := \mathbb{E}_{x \sim q(x)} \log\left(\frac{q(x)}{p(x)}\right)$. Recall the ELBO for variational diffusion model is defined as

$$E(p; q) := \mathbb{E}_{q(x_0)} \mathbb{E}_{q(x_{1:T} | x_0)} \log \frac{p(x_{0:T})}{q(x_{1:T} | x_0)}.$$

- i. (1pt) Generative diffusion modeling aims to generate samples whose distribution is close to that of an observed set of samples. Show that the KL divergence $D_{\text{KL}}(q(x_{0:T}) \parallel p(x_{0:T}))$ between the forward/backward processes can be expressed in terms of ELBO,

$$D_{\text{KL}}(q(x_{0:T}) \parallel p(x_{0:T})) = -E(p; q) + \mathbb{E}_{q(x_0)} [\log q(x_0)].$$

Solution:

- ii. (1pt) Generative diffusion modeling alternatively works as maximizing the log-likelihood of an observation generated by the diffusion model. Show that the log-likelihood $\mathbb{E}_{q(x_0)} [\log p(x_0)]$ can be expressed in terms of ELBO,

$$\mathbb{E}_{q(x_0)} [\log p(x_0)] = E(p; q) + \mathbb{E}_{q(x_0)} [D_{\text{KL}}(q(x_{1:T} \mid x_0) \parallel p(x_{1:T} \mid x_0))].$$

Solution:

- iii. (1pt) Show that the ELBO maximization and the KL divergence minimization are equivalent, i.e., their solutions are the same,

$$\operatorname{argmax}_p E(p; q) = \operatorname{argmin}_p D_{\text{KL}}(q(x_{0:T}) \parallel p(x_{0:T})).$$

Solution:

- iv. (2pts) Moreover, show that a solution p^* of the problems in (iii) is a solution for the log-likelihood maximization problem, i.e.,

$$p^* \in \operatorname{argmax}_p \mathbb{E}_{q(x_0)} [\log p(x_0)].$$

Solution: